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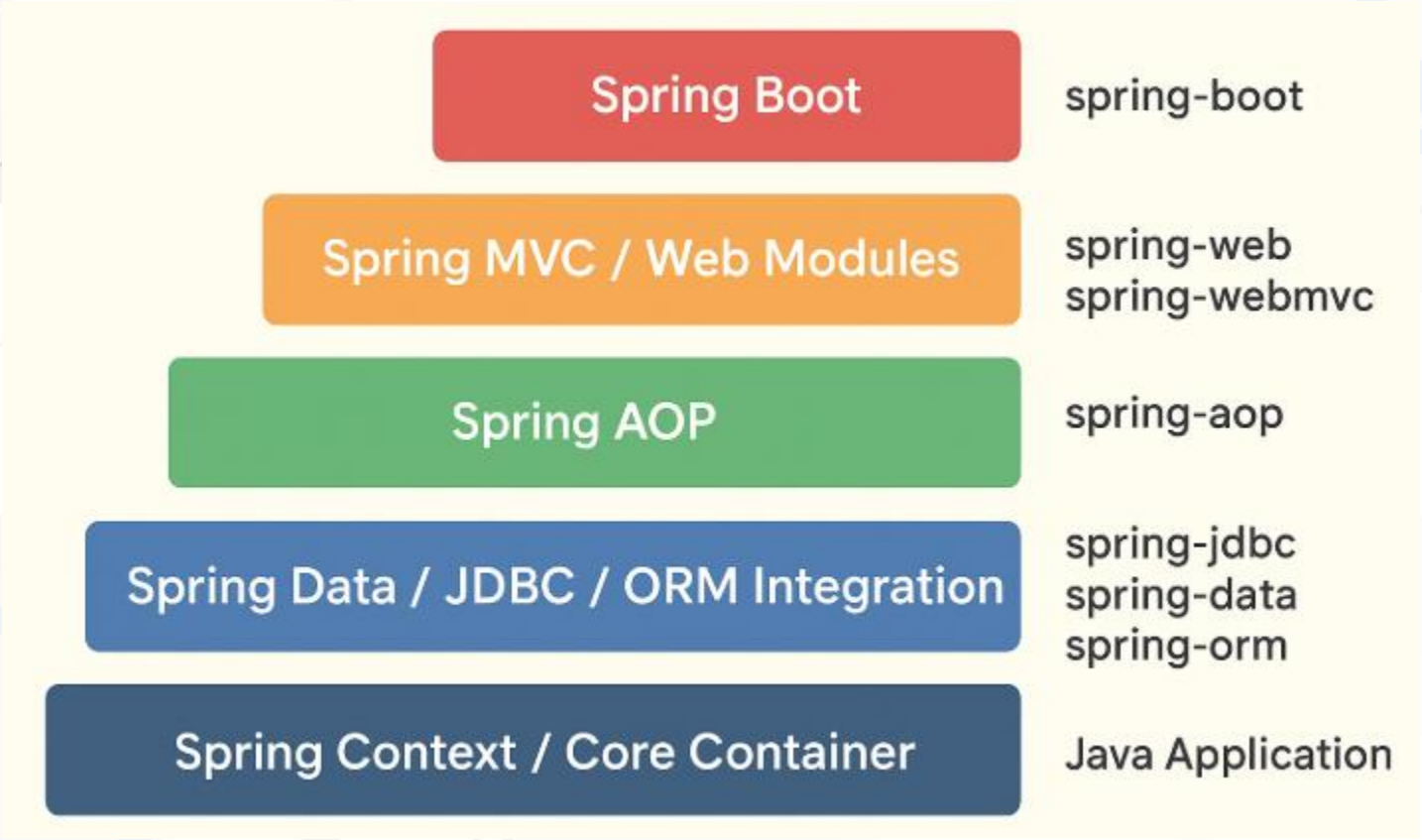
WELCOME

Introduction

- Spring Framework is a **powerful and lightweight** Java framework used to build **enterprise-level** applications easily and efficiently.

- Spring was introduced to simplify Java development by handling all those complex parts like object creation, dependency management, transactions, and security automatically.

Spring Architecture Diagram



Inversion of Control (IOC)

- Inversion of Control (IoC) is a principle in software engineering where the control of object creation and management is transferred (or inverted) from the developer to the framework

Why IOC ?

- ✓ **Loose Coupling** — classes don't depend directly on concrete implementations
- ✓ **Easier Testing** — objects can be replaced with mock dependencies
- ✓ **Better Code Management** — Spring handles the lifecycle of beans automatically

Two main container types

BeanFactory

ApplicationContext

BeanFactory

- BeanFactory is the simplest container in Spring that provides basic IoC (Inversion of Control) and Dependency Injection support.
- Creates beans **lazily** (only when requested)
- Lightweight
- Doesn't support advanced features like event publishing or internationalization

```
BeanFactory factory = new XmlBeanFactory(new ClassPathResource("applicationContext.xml"));
Student s = (Student) factory.getBean("s1");
s.display();
```

ApplicationContext

- ApplicationContext is a more advanced container that extends BeanFactory and provides additional enterprise-level functionality.

```
ApplicationContext context = new ClassPathXmlApplicationContext("applicationContext.xml");
Student s = context.getBean("s1");
s.display();
```

Steps To Create Spring Application

1. Create the Java Project.
2. Add spring jar files
3. Create Java class
4. Create the xml file
5. Create the Main method class

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Steps

1. Create a new **Java Project**

2. Create bean/model file in your package like **Demo/Student.java**

```
package Demo;

public class Student {
    private String name;
    private String address;

    //generate getter & setters

    public void displayinfo() {
        System.out.println(name+" "+address);}
    }
}
```

3. Create **applicationContext.xml** file in **src** package

```
<?xml version="1.0" encoding="UTF-8"?>
<beans
      xmlns="http://www.springframework.org/schema/beans"
      xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
      xmlns:p="http://www.springframework.org/schema/p"
      xsi:schemaLocation="http://www.springframework.org/schema/beans
      http://www.springframework.org/schema/beans/spring-beans-3.0.xsd">

       <bean id="s1" class="Demo.Student">
              <property name="name" value="Abcd "></property>
              <property name="address" value="Pune "></property>
       </bean>

</beans>
```

4. Create **Test.java** with **main method**

```
ApplicationContext context = new ClassPathXmlApplicationContext("applicationContext.xml");

Student s = (Student) context.getBean("s1");
s.displayinfo();
```

OR

```
BeanFactory factory = new XmlBeanFactory(new ClassPathResource("applicationContext.xml"));
Student s = (Student) factory.getBean("s1");
s.display();
```

Dependency Injection (DI)

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- Dependency Injection (DI) is a design pattern used by the Spring Framework to automatically provide (inject) the required objects (dependencies) into a class, instead of creating them manually.

- When **one class depends on another class**, that's a *dependency*

Types of Dependency Injection

1. Constructor Injection
2. Setter Injection

Setter Injection

- Setter Injection is a type of Dependency Injection where the Spring container injects dependencies using setter methods of a bean (class).

- Spring calls the setter methods after creating the bean to supply the required dependencies.

Student.java

```java
private String name;
private String address;
```

```java
public String getName() {
        return name;
}

public void setName(String name) {
        this.name = name;
}

public String getAddress() {
        return address;
}
public void setAddress(String address) {
        this.address = address;
}
```

Xml file

```
<bean id="s1" class="Demo.Student">
        <property name="name" value="abcd"></property>
        <property name="address" value="pune"></property>
</bean>
```

Constructor Injection

- Constructor Injection is a type of Dependency Injection (DI) where the Spring container injects dependencies through a class constructor.

- When Spring creates the object (bean), it passes the required dependencies as constructor parameters.

Student.java

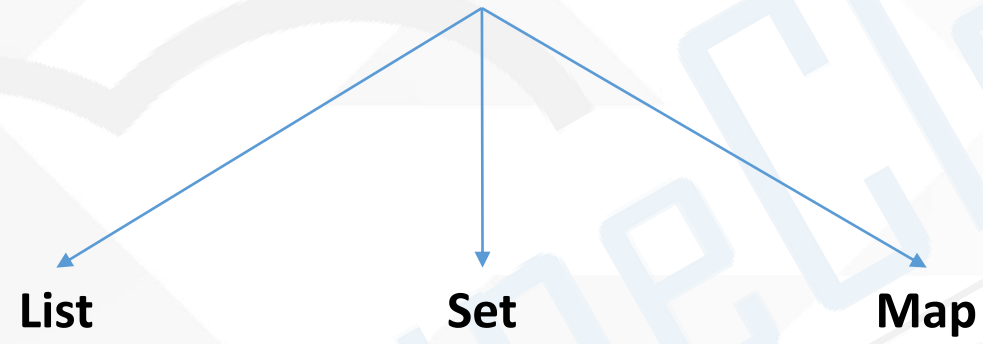
```
private int id;
private String name;
```

```
public Student() {System.out.println("Default Constructor");}

//single parameter constructor
public Student(int id) {this.id=id;}
public Student(String name) {this.name=name;}

public Student(int id, String name) {
        this.id=id;
        this.name=name;
}
```

Xml file

```
<bean id="e" class="Demo.Student">
        <constructor-arg value="10" type="int"></constructor-arg>
        <constructor-arg value="abc"></constructor-arg>
</bean>
```

Constructor Injection with Collection



Constructor Injection with Collection - List

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Student.java //step 1

```
public List<String> skills;
```

Student.java //step 2

```
System.out.println(skills);
```

OR

```
Iterator<String> itr = skills.iterator();

while(itr.hasNext()) {
        System.out.println(itr.next());
}
```

Xml file

```
<constructor-arg>
        <list>
                <value>Java</value>
                <value>Python</value>
        </list>
</constructor-arg>
```

Constructor Injection with Collection - Set

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Student.java //step 1

```
private Set<String> departments;
```

Student.java //step 2

```
System.out.println(departments);
```

OR

```
Iterator<String> itr = departments.iterator();

while(itr.hasNext()) {
        System.out.println(itr.next());
}
```

Xml file

```
<constructor-arg>
        <set>
                <value>Computer</value>
                <value>Mechanical</value>
        </set>
</constructor-arg>
```

Constructor Injection with Collection - Map

Xml file

Student.java //step 1

```java
private Map<String, String> books;
```

```xml
<constructor-arg>
    <map>
        <entry key="Book 1" value="Author 1"/>
        <entry key="Book 2" value="Author 2"/>
    </map>
</constructor-arg>
```

Student.java //step 2

```java
System.out.println(books);
```

OR

```java
Iterator<Map.Entry<String, String>> itr2 =
books.entrySet().iterator();

while (itr2.hasNext()) {
  Map.Entry<String, String> e = itr2.next();
  System.out.println(e.getKey() + " " + e.getValue());
}
```